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CEO Incentives and Customer-Supplier Relations

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This study explores how suppliers adjust their relation-specific investments (RSI) in response to the different risk-taking incentives provided by the customer firm to its CEO, during normal and transition periods. We investigate this relation using 17,553 customer–supplier transactions over the 1993–2013 period. We find strong evidence consistent with the risk-taking argument. Specifically, we find that an increase in the risk-taking incentives of customer CEOs leads to a decline in suppliers' RSI in normal periods, but an increase in RSI during transition periods. We employ the FAS-123R mandate to show that an *exogenous* reduction in customer CEO's incentive pay increases suppliers' RSI. We reaffirm the effect with the passage of the Sarbanes–Oxley Act as a secondary quasi-natural experiment. Finally, we examine several scenarios that either amplify or attenuate the observed relation, based on factors such as financial constraints, distress, growth opportunities, industry competition, and other firm characteristics. Our study contributes to the literature that examines the interplay between corporate policy and product market relationships.

Key words: risk-taking; incentives; executive compensation; option grants; supply chain
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1. Introduction

Among firms that are economically linked (such as customers and suppliers), one firm's decisions will likely have a material impact on the other. Prior literature documents several instances of the interplay between a firm's decisions and product market relationships. For instance, decisions regarding a customer firm's leverage (Banerjee et al. 2008, Kale and Shahrur 2007) or earnings management (Raman and Shahrur 2008) are shown to be related to its suppliers' relation-specific investments (RSI hereafter). In this study, we focus on the relation between a customer firm's CEO compensation and its suppliers' RSI. Specifically, we examine how suppliers adjust their RSI in response to the customer firm's CEO option grants, during periods when customers experience a demand or productivity shock (hereafter referred to as transition periods). We refer to non-transition periods as normal periods.

Broadly, RSI are investments made by a supplier that are specific to a customer firm. The well-documented success of Toyota and Honda, for example, is in large part due to their reliance on suppliers who have made investments in acquiring compatible assets (Chen and Deng 2015, Liker and Choi 2004). We focus on suppliers' RSI because such investments are risky due to high switching costs. Consequently, RSI typically have a higher value within the customer–supplier (C-S hereafter) relationship than outside it and whose payoff depends on the relationship's continuation (Crawford 1990).

To differentiate between C-S relations during normal and transition periods, we follow Ahern and Harford (2014) and Harford (2005) to define shocks (and hence transition and normal periods). Briefly, we use a principal component analysis to create an index that aggregates year-to-year changes in seven fundamental variables that affect the demand and productivity of firms in an industry. A shock (or transition period) occurs when the value of this aggregated change index lies in the top quartile of its distribution. Intuitively, transitional periods occur when a firm's industry experiences significant changes in key fundamental characteristics that are likely to impact its demand and/or productivity. All other periods are classified as normal periods.

In general, we expect a negative relationship between option grants to the customer's CEO and its suppliers' RSI, because option grants provide incentives for the customer to take on additional risk. To bolster confidence in this risk-taking argument along the supply chain, we use shocks (or transition periods) as an additional treatment. Our results show that

during normal periods, suppliers decrease RSI when the customer firm's CEO receives new option grants. However, suppliers increase RSI when CEO options are granted during a transition period, that is, when there is a shock to the customer's industry. These results are consistent with a risk-taking argument. Essentially, the risk-taking hypothesis argues that CEO option grants incentivize the customer CEO to undertake risky projects and hence increase the firm's cash flow volatility. During normal periods, undertaking risky projects increases the likelihood of financial distress, which undermines the value of suppliers' RSI. Therefore, suppliers invest less in RSI. However, during transition periods, taking on increased risks may enhance the value of suppliers' RSI. This can happen either because the customer firm may be distressed and wants to "gamble" its way out of trouble or, because it wants to invest in riskier but higher payoff projects to fully take advantage of the shock. Accordingly, suppliers invest more in RSI during transition periods, but invest less during normal periods. We refer to this hypothesis as *risk-taking* and develop it in section 2.

We use the *Compustat Historical Segment* database to obtain data on customer–supplier pairs from 1993 to 2013. A supplier can serve multiple customers while a customer firm can transact with multiple suppliers. To mitigate potentially confounding events for different customers (such as one experiences a shock, and another does not) and to account for heterogeneity in unobserved C-S pair characteristics, we conduct all our analyses with C-S-year as the unit of observation and include C-S pair level fixed effects in all specifications. This setup treats each transaction as an independent observation and lends itself to a cleaner and more robust interpretation of the findings.

To further identify the relation between customer firms' CEO option grants and suppliers RSI, we conduct two quasi-natural experiments that employ a difference-in-difference (DID) design. The first of these utilizes the 2005 mandate requiring firms to comply with FAS-123R. This federal regulation required firms to expense option grants at their fair value which caused a significant cutback in option pay thereby reducing the sensitivity of CEOs' wealth to stock volatility (or vega) (e.g., Bakke et al. 2016, Chava and Purnanandam 2010, Hayes et al. 2012). First, we find that customer firms experiencing an exogenous decline in option grants due to FAS-123R saw a corresponding increase in supplier RSI. However, if we only focus on customer firms during transition periods, we find a significant reduction in suppliers' RSI for these firms after they experience an exogenous decline in option grants. This test provides further evidence that *exogenously* induced changes in the risk-taking incentives of customer-firm CEOs lead their suppliers to adjust their RSIs. In particular, it helps tease out alternative explanations that other hidden variables cause the seeming correlation between CEO option grants and suppliers' RSI.

As an additional check, we undertake a second DID experiment, using the passage of the Sarbanes–Oxley Act (SOX) in 2002. To the extent that SOX increases legal/political exposure and litigation risk, it should be associated with a reduction in risk-taking (Cohen et al. 2013). Results from this test reinforce our findings from the FAS-123R experiment. There is a significant reduction in post-SOX supplier RSI for customer firms that are affected by the regulation relative to the control sample of firms that are unaffected by SOX, during transition periods. We conduct multiple placebo tests that accompany both DID tests. These include randomizing the event or "shock" year of the regulation to pre- and post-periods as well as randomizing the control sample. We find that our results hold only around the actual event and not around any of the placebo events. As with the baseline tests, we conduct all the DID analyses, using transaction level data with C–S pair fixed effects.

Next, we decompose the overall shock into demand and productivity components and further classify these into positive and negative shocks. We find that suppliers on average increase RSI, following an option grant coupled with a *negative shock*. These results provide additional support for the risk-taking hypothesis. Finally, we undertake several additional tests to reaffirm our findings. Specifically, we find that our results manifest in the following subsamples: (i) customer firm is financially constrained but is not in economic distress, (ii) supplier is financially constrained or in economic distress, (iii) customer firm faces low growth opportunities and/or operates in a competitive industry, and (iv) supplier faces high growth opportunities and/or operates in a competitive industry. Additionally, we find that high R&D supply chains exhibit greater risk-taking than low R&D supply chains. These scenarios represent situations wherein the supplier is most likely to increase risk or RSI when a customer firm faces a shock and increases its own risk-taking incentives. They offer additional evidence for an increase in risk-taking along the entire supply chain when a customer experiences a shock. We also rule out certain alternate explanations that pertain to earnings management and merger involvement by the transacting parties. Finally, we conduct placebo tests that involve cases with multiple customers (or industries), government, and family firms as customers. Findings from these tests provide additional insights discussed in detail in section 5.

Our study makes two principal contributions. First, to our knowledge, we are the first to document the effect of customer firms' CEO incentives on suppliers'

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RSI, using natural experiments. Our findings contribute to the growing body of empirical research that examines the interplay between corporate risk policy and product market relationships. For instance, firms reduce leverage to induce higher RSI from their suppliers and customers (Banerjee et al. 2008, Kale and Shahrur 2007) or warranty offering firms increase (lower) debt and therefore financial risk when distressed (non-distress) (Kale et al. 2013). Our evidence of risk-taking along the supply chain is consistent with these studies.

Our second contribution is to the literature that examines spillover effects along the supply chain. In particular, Fee et al. (2006) document that factors related to contractual incompleteness are important determinants of a firm's decision to take an equity stake in their suppliers. Radhakrishnan et al. (2014) find that the quality of customers' information has a spillover effect on the operating performance of their upstream supplier firms while Pandit et al. (2011) examine the reaction of suppliers' stock prices to their customers' earnings announcements. Wagner et al. (2012) document a positive association between supply chain fit and financial performance. Modi and Mabert (2007) develop a conceptual model and use survey data to show that in order to remain competitive, customers expend considerable resources to improve supplier performance through knowledge transfer. Hendricks and Singhal (2005) find that disruptions in the supply chain not only have an adverse effect on suppliers but also create uncertainties. In a recent study, Osadchiy et al. (2016) examine channels through which systematic risk propagates along the supply chain. We complement this stream of literature by investigating the spillover effect of a shock in the customer's industry on suppliers' RSI.

Finally, in a recent commentary on the interplay between operations and finance, Birge (2015) calls for work on "…the role of incentives and financial market activity in operational decision making." Our study addresses this gap and adds to the body of work that examines the relation between a firm's CEO incentives and features of its product input markets.

2. Background and Hypothesis Development

This study examines customer–supplier relations based on the underlying premise that complete contracts are prohibitively costly since it is difficult for the transacting firm to foresee all future contingencies and contract upon them (Williamson 1981). Consequently, C–S relations are typically characterized by a nexus of implicit and explicit but incomplete contracts (Milgrom and Roberts 1992). In such a contracting environment, information on a customer firm's future prospects and policies is especially important for suppliers to manage their risk during transitional periods or in the presence of high uncertainty. Transitional periods often represent major strategic shifts in firm policy. For instance, Ahern and Harford (2014) argue that shocks to an industry are transmitted through the complex C-S network and affect other economically linked industries. They provide compelling evidence to show how this induces major strategic initiatives such as merger waves. The authors further contend that reorganization of a supplier (or customer) industry could cause a customer (supplier) industry to reorganize in response. In a similar vein, Osadchiy et al. (2016) map supply networks of firms and industries to investigate how systematic risk propagates through this network while Kesavan and Kushwaha (2014) show that macroeconomic shocks impact the inventory investment behavior of retailers. We measure uncertainty or transition periods as a shock to fundamental factors in the customer firm's industry and study the ensuing effects on its suppliers' RSI. Finally, Cohen and Frazzini (2008) examine return predictability across the supply chain and explicitly state "Thus, any shock to one firm has a resulting effect on its linked partner." We look for heterogeneity in customer firms and the consequent responses in their suppliers' RSIs. To this end, we develop the *risk-taking* hypothesis below.

Our motivation for the risk-taking hypothesis in large part stems from the widely documented view that the value of option grants increases with the firm's volatility (or vega). Option compensation therefore creates an incentive to take on more risk (Coles et al. 2006). This would lead to increased cash flow volatility, consistent with the findings of Shue and Townsend (2016) who document that a 10% increase in new options granted leads to a 2.8% to 4.2% increase in equity volatility. Further, Campbell et al. (2008) show that an increase in equity volatility leads to a significant increase in the firm's predicted *long-term* failure probability. Thus, during normal periods, an increase in market volatility will potentially undermine the value of RSI and consequently lead suppliers to lower RSI. This is also consistent with prior studies. For instance, firms reduce leverage (or risk) to induce higher RSI from their suppliers and customers (Banerjee et al. 2008, Kale and Shahrur 2007).

However, when the customer firm experiences a shock, an increase in risk could propel a distressed firm out of trouble, by increasing its probability of exiting the distressed state. For instance, a customer firm may have "little to lose but a lot to gain" by taking on a risky high-payoff project when in distress (e.g., Jensen and Meckling 1976). For a distressed firm, such a project would have a low downside with a

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potentially high upside. In other words, customers may want to "gamble" their way out of trouble using the convexity of the risky project's payoff. Suppliers may support this endeavor consistent with Petersen and Rajan (1997) who find that suppliers increase trade credit to customer firms with negative sales growth. Alternately, a shock to the customer industry may also open up opportunities (Kogan and Papanikolaou 2014), such as the discovery of new technology. In this case, a customer firm may choose to invest in this opportunity either to remain competitive or enhance growth. Consequently, the customer firm may provide their managers with greater risk-taking incentives to pursue these risky but high payoff strategic opportunities either to exit from a distressed state or to fully capitalize on an opportunity. In either scenario, the value of suppliers' RSI is potentially enhanced. This would lead suppliers to increase RSI when risk-taking incentives to the customer firm accompany a shock. Kale et al. (2013) provide a similar intuition—warranty offering firms increase (lower) debt and therefore financial risk when distressed (non-distress). The above arguments lead to the *risk-taking* hypothesis stated below:

Risk-taking Hypothesis:

H1. *During normal periods, customer firms' CEO option grants are negatively associated with suppliers' RSI.*

H2. *Compared to normal periods, customer firms' CEO option grants are less negatively associated with suppliers' RSI during transition periods.*

3. Sample Construction, Variable Definitions, and Empirical Methodology

3.1. Customers, Suppliers, and Transactions

Following the mandate by *The Statement of Financial Accounting Standards* No. 14 (SFAS 14), firms are required to disclose the identities (names) of key customers, i.e., those that account for at least 10% of their total revenue or are deemed important to their business. We follow prior literature (e.g., Fee and Thomas 2004, Kale and Shahrur 2007, Radhakrishnan et al. 2014) and begin with approximately 29,000 C-S pairs from the *Compustat Segment Files*. We then merge this initial sample with customer compensation data from *ExecuComp* over the period 1993–2013. We restrict our sample to public firms with at least two consecutive years of customer- and supplier-related information and exclude firms in the financial sector (SIC 6000–6990). Our final sample consists of 17,553 C–S-year transactions. There are 4,333 unique customer-year and 12,822 supplier-year observations.[1] We conduct our analyses using disaggregated data at the C-S-year level. Panel A of Table 1 reports details of the sample construction.

Panel B of Table 1 presents key summary statistics. The average customer firm in our sample buys $234 million worth of goods from its supplier, which accounts for about 1.4% of its *Cost of Goods Sold*. In contrast, the average (median) transaction with *each* customer accounts for 18.9% (14.3%) of a supplier's annual sales. A typical C-S pair has transacted with each other for 6 years. The average firm buys from about four suppliers while the average supplier reports 1.4 customers. Customers are considerably larger than suppliers with average annual sales of $18.5 billion compared to $1.3 billion for suppliers.

3.2. Suppliers' Relationship-specific Investment

The objective of our main dependent variable is to proxy for suppliers' investment in the C-S relation. Consequently, we use the annual transaction weighted R&D intensity for each C-S pair as the proxy for suppliers' RSI in the customer firm, where the weights represent the importance of the particular customer to the corresponding supplier. We construct the variable *Supp. RSI* in keeping with extant literature (e.g., Kale and Shahrur 2007, Raman and Shahrur 2008). Several studies illustrate the link between R&D and asset specificity. For instance, Allen and Phillips (2000) and Acemoglu et al. (2010) contend that R&D-intensive industries are more likely to create relationship-specific assets. Armour and Teece (1980) argue that R&D intensive firms along the vertical chain have complex inter-dependencies while Levy (1985) argues that these firms use specialized inputs that require transaction-specific investments by suppliers. Phillips and Zhadanov (2013) claim that large firms can optimally outsource their R&D to smaller firms and subsequently acquire them upon successful innovation, since their R&D investments are potentially tailored to the firm's inputs. Bowen et al. (1995) and Titman (1984) use a firm's R&D expenditure as an indicator of product uniqueness (or asset specificity). Together, this body of work suggests that R&D intensity of suppliers serves as a reasonable proxy for suppliers' RSI. Consequently, we construct the variable *Supp. RSI* using the *Cust. Importance* weighted R&D as follows:

$$Supp.\ RSI_{i,j,t} = Cust.\ Importance_{i,j,t} \times Supp.\ R\&D_{i,t} \quad (1)$$

where:

$$Supp.\ R\&D_{i,t} = (\text{Supplier } i\text{'s R\&D})_{i,t}/(\text{Supplier } i\text{'s Total Assets})_{i,t} \quad (2)$$

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Table 1 Sample Construction and Descriptive Statistics

Panel A: Sample and filter	Customer-supplier year	Customer year	Supplier year
Raw *Compustat Segment* files w/identifier	28,817		
Merge with customer info. on *ExecuComp*	23,575	6,109	16,675
Drop missing customer variables (t)	23,136	5,929	16,515
Drop missing supplier variables (t and $t+1$)	17,553	4,333	12,822

Panel B: Descriptive statistics	No. observations	Mean	25th percentile	Median	75th percentile	SD
Customer–supplier pairs						
Transaction ($ million)	17,553	233.52	7.16	26.07	106.49	1,070
Customer Importance (Cimp)	17,553	0.189	0.10	0.143	0.23	0.163
Supplier Importance (Simp)	17,553	0.014	0.00	0.002	0.008	0.056
C-S Years	17,553	6.6	3	6	9	4
Customer compensation						
ST Comp. ($ 000s)	4,333	2,507	1,093	1,896	3,131	2,219
LT Comp. ($ 000s)	4,333	5,765	1,108	3,324	7,656	7,788
Tot Comp. ($ 000s)	4,333	8,315	2,666	5,561	10,752	8,911
Option grants (% of Tot Comp.)	4,333	32.5%	0.0%	29.3%	52.9%	28.2%
Stock awards (% of Tot Comp.)	4,333	16.5%	0.0%	0.0%	31.4%	22.6%
Bonus (% of Tot Comp.)	4,333	13.2%	0.0%	8.1%	22.4%	15.7%
ST Comp (% of Tot Comp.)	4,333	44.0%	24.8%	38.7%	58.0%	25.9%
Customer characteristics						
No. suppliers	4,333	4.05	1.00	2.00	4.00	8.200
Customer Sales ($ million)	4,333	18,490	2,414	7,486	18,606	35,572
Customer Assets ($ million)	4,333	20,180	2,236	6,794	19,844	41,254
Customer Book Leverage	4,333	0.236	0.115	0.225	0.335	0.159
Customer Book to market	4,333	0.614	0.418	0.62	0.798	0.247
Customer ROA	4,333	0.053	0.026	0.055	0.091	0.074
Customer SD returns	4,333	0.101	0.062	0.087	0.125	0.054
Customer R&D	4,333	0.030	0	0	0.046	0.047
Supplier characteristics						
No. customers	12,822	1.37	1	1	1	0.910
Supplier sales ($ million)	12,822	1,346	51	187	786	3,820
Supplier assets ($ million)	12,822	1,568	54	215	1,012	4,152
Supplier book leverage	12,822	0.24	0.02	0.19	0.38	0.24
Supplier R&D	12,822	0.07	0	0.01	0.09	0.12

Notes: The sample consists of 17,553 observations, spanning the 1993–2013 period, with complete information on customer compensation and data on customer and supplier characteristics. Panel A of the table reports details on the sample construction and panel B presents descriptive statistics on key variables. All variables are defined in the *Data Appendix*.

$$Cust.\,Importance_{i,j,t} = (\text{Supplier } i\text{'s Sales to Cust. } j)_{i,j,t} / (\text{Supplier } i\text{'s Total Sale})_{i,t} \tag{3}$$

where t indexes time; j refers to customer firm j; i refers to supplier i to customer j. The average supplier invests 7% of their assets toward R&D each year. The variable *Cust. Importance* is essentially a measure of the relative importance of each major customer for a supplier's annual revenue and used to construct measures of customer concentration. Recent studies show that customer concentration is positively associated with a supplier's accounting and stock price performance measures (Patatoukas 2012). In a related study, Osadchiy et al. (2016) quantify the impact of changes in customer-base concentration on the firm's systematic risk and risk-adjusted valuation. The dependent variable *Supp.* $RSI_{i,j,t,t+1}$ is the sum of *Supp.* $RSI_{i,j,t}$ and *Supp.* $RSI_{i,j,t+1}$. This effectively ensures that all independent variables are lagged by at least 1 year and adhere to a timeline that allows suppliers to adjust their R&D investments after observing the shock to the customer's industry. Since our analyses are at the disaggregated level, a customer with n suppliers in a given year will have n individual transactions (or observations). To account for the resulting repetition in the customer-level variables (such as shock) in this set-up, we cluster standard errors at the customer firm and year levels in *all* our specifications.

3.3. CEO Option Grants

We obtain option grants data from *ExecuComp*. The variable *Option Grants* is the ratio of the annual incremental option grants to the CEO's total annual compensation. The total value of annual option grants is provided by *ExecuComp* and is computed using the

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Black–Scholes value. Because a CEO can receive multiple grants in a year, the annual incremental value of option grants is the sum of fiscal-year-end values of all options granted during the year. The average CEO's annual total compensation is about \$8.3 million, of which \$2.5 (\$5.8) million is short-term (long-term) compensation.[2] The average value of *Option Grants* is 32.5%. Because any new grants will likely depend on a CEO's firm-specific equity portfolio at the beginning of the year, we compute the dollar sensitivity of the CEO's accumulated stock and option portfolio to a 1% change in the firm's stock price (or delta) as the natural log of (1 + delta) in the CEO's firm-specific equity portfolio. To account for the change in reporting format mandated by FAS-123R (Coles et al. 2014), we include a binary indicator to denote whether the firm's compensation is based on the old/new reporting format. Our results are robust to the exclusion of delta as well as observations with a switch in reporting format.

3.4. Economic Shocks

We follow the methodology in Harford (2005) and Ahern and Harford (2014) to construct a proxy for economic shocks. First, we compute the median absolute one-year change in each of the following seven variables at the 2-digit SIC industry-year level: net income to sales (NI), asset turnover (ATO), ROA, R&D to assets (RND), capital expenditures to assets (CAPX), employee growth rate (EGR), and sales growth rate (SGR). Because these variables are highly correlated with each other within an industry, resulting in multi-collinearity if simultaneously included in a regression model, we use an eigen value decomposition to produce a set of principal components that are linear combinations of the seven underlying variables. We extract the first principal component which explains the largest possible variance in the underlying variables, for each 2-digit SIC industry-year. We then define *Cust. Shock* (*Supp. Shock*) as a binary indicator that equals one if this principal component for the customer's (supplier's) industry-year is in the top quartile, and zero otherwise (Ahern and Harford 2014). We further classify this overall shock measure into productivity (NI, ATO, ROA) and demand (RND, CAPX, EGR, SGR) components. Next, we define a positive (negative) demand and productivity shock if the median changes in *all* the underlying variables *within* each category are positive (non-positive). The intuition behind this approach is that a shock represents the historically largest movements (in either direction) in a composite index of the seven underlying fundamental economic factors listed above. Ahern and Harford (2014) demonstrate the validity of this measure by relating it to the incidence and timing of merger waves across customer-supplier industry pairs from 1986 to 2010.

3.5. Control Variables

We employ several controls for the customer and supplier. Nearly 69% of CEOs also hold the position of *Chairman* of the Board while about 8.3% are in their first year at the helm. Consistent with similar studies (e.g., Kale and Shahrur 2007), we include accounting- and market-based measures such as *Size*, *ROA*, *Book to market*, standard deviation of returns, leverage, and the sales-based herfindahl index to control for firm performance, growth opportunities (Aggarwal and Samwick 1999), and industry competition, respectively. Finally, we include the customer firm's own R&D intensity to control for the likelihood that innovative firms rely on R&D intensive suppliers. To account for aspects that pertain to suppliers we include analogous controls for each supplier. Two important supplier level controls are noteworthy. First, to control for shocks to the supplier's own industry, we include the variable *Supplier Shock* which is constructed in exactly the same manner as *Customer Shock*. The second variable is the binary indicator *Single Industry Supplier*, which equals one if all the customers for a given supplier in a year are in the same industry, and zero otherwise. This variable controls for temporal changes in a supplier's customer industry base and for suppliers with customers in multiple industries and potentially experiencing differential shocks in a year.

3.6 Empirical Methodology

We test the risk-taking hypotheses using the baseline specification in Equation (4) below:

$$\begin{aligned} Supp.\,RSI_{t,t+1} = {} & B_0 + B_1 Option\ Grants \\ & + B_2 Cust.\,Shock + B_3 Option\ Grants \\ & \times Cust.\,Shock + B_4 Supp.\,Shock \\ & + B_5 Cust.\,Controls + B_6 Supp.\,Controls \\ & + C\text{-}S\ FE + Year\ FE + \mathrm{e}_t \end{aligned} \tag{4}$$

All specifications include C-S pair fixed effects (FE) to account for heterogeneity across C-S pairs and year FE to account for temporal variations in the macroeconomic environment. We winsorize all continuous variables at the 1% tails. On average, a customer firm has about four suppliers in a given year. About 46% of customer firms have a single supplier in a year, 64% have at most two, and nearly 93% have fewer than ten suppliers annually. Thus, there are repeats of customer-year specific observations depending on the number of suppliers for each customer. Consequently, we compute *t*-statistics based on robust

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Table 2 Customers' Compensation and Suppliers' Relationship-Specific Investment (RSI)

	(1) *Supp. $RSI_{t,t+1}$*	(2) *Supp. $RSI_{t,t+1}$*	(3) *Supp. $RSI_{t,t+1}$*	(4) *Supp. $RSI_{t,t+1}$*
Option grants	−0.0037*** (−2.89)	−0.0037*** (−2.89)	−0.0062*** (−4.22)	−0.0062*** (−4.24)
Customer shock		−0.0002 (−0.20)	−0.0032** (−2.09)	−0.0036** (−2.36)
Option grants × Customer shock			0.0074** (2.54)	0.0072** (2.47)
Customer characteristics				
New CEO	0.0003 (0.30)	0.0003 (0.30)	0.0004 (0.33)	0.0004 (0.35)
Chairman	−0.0014 (−1.32)	−0.0014 (−1.32)	−0.0013 (−1.25)	−0.0013 (−1.21)
Delta	−0.0003 (−1.22)	−0.0003 (−1.21)	−0.0003 (−1.21)	−0.0003 (−1.17)
Old format	0.0003 (0.13)	0.0003 (0.13)	0.0006 (0.25)	0.0005 (0.22)
Customer size	0.0018 (0.98)	0.0018 (0.97)	0.0017 (0.90)	0.0017 (0.91)
Customer book leverage	0.0060 (1.17)	0.0059 (1.16)	0.0062 (1.21)	0.0061 (1.19)
Customer book to market	0.0002 (0.09)	0.0002 (0.08)	0.0001 (0.02)	0.0001 (0.02)
Customer ROA	0.0182* (1.86)	0.0183* (1.86)	0.0185* (1.88)	0.0179* (1.82)
Customer SD returns	0.0019 (0.20)	0.0020 (0.22)	0.0023 (0.25)	0.0019 (0.21)
Customer R&D	0.0751 (1.45)	0.0749 (1.44)	0.0752 (1.45)	0.0748 (1.44)
Missing R&D	−0.0049*** (−3.89)	−0.0049*** (−3.89)	−0.0049*** (−3.87)	−0.0044*** (−3.58)
Customer Ind. HHI	0.0112** (2.51)	0.0113** (2.52)	0.0114** (2.53)	0.0111** (2.48)
Supplier characteristics				
Supplier shock				0.0025** (2.08)
Supplier size	−0.0038*** (−2.66)	−0.0038*** (−2.66)	−0.0037*** (−2.64)	−0.0038*** (−2.66)
Supplier book leverage	−0.0024 (−0.68)	−0.0024 (−0.69)	−0.0023 (−0.68)	−0.0023 (−0.65)
Supplier ROA	−0.0236*** (−6.64)	−0.0236*** (−6.65)	−0.0234*** (−6.62)	−0.0234*** (−6.62)
Supplier Ind. HHI	−0.0010 (−0.33)	−0.0010 (−0.33)	−0.0012 (−0.39)	−0.0014 (−0.45)
Single customer dummy	0.0032*** (2.92)	0.0032*** (2.91)	0.0032*** (2.89)	0.0031*** (2.80)
Customer–Supplier pair FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Adjusted R^2	0.808	0.808	0.808	0.808
Observations	17,553	17,553	17,553	17,553

Notes: The table reports the results of estimating Equation (4). The constant is suppressed un-reported. The sample contains 17,553 customer-supplier-year observations, from 1993 to 2013. The dependent variable is *Supp. $RSI_{t,t+1}$*. All regressions include C-S pair and year fixed effects. The *t*-statistics are based on standard errors clustered by customer-firm and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in the *Data Appendix*.

standard errors that are clustered by customer firm and year in all our analyses.

4. Discussion of Results

4.1. Baseline Analysis

We begin with the baseline specification in Equation (4) to examine the relation between customer CEO compensation, economic shocks, and supplier RSI (see Table 2). The first column includes only *Option Grants* as the main compensation-related independent variable. The estimated coefficient on *Option Grants* is negative (= −0.0037) and statistically significant at the 1% level (*t*-stat = −2.89). This shows that suppliers on average react negatively to risky option compensation and reduce their RSI. The next specification includes *Customer Shock* as an additional variable. The estimate on *Option Grants* continues to remain negative and significant, but *Customer Shock* on average has no effect on supplier RSI. The third specification goes a step further and includes the interaction term *Option Grants* × *Customer Shock*. The addition of the interaction term reveals interesting insights. First, the negative coefficient on *Option Grants* continues to remain negative. The magnitude of the estimate, however, is now much larger (increases from −0.0037 to −0.0062) with higher statistical significance (*t*-stat increases from −2.89 to −4.22). Given that option grants are riskier (Coles et al. 2006), this evidence is consistent with Kale and Shahrur (2007) who find that suppliers' RSI is negatively related to customers' debt levels (or financial risk). Further, the interaction term *Option Grants* × *Customer Shock* is positive (=0.0074) and statistically significant at the 5% level (*t*-stat = 2.54). These findings are consistent with the risk-taking argument because more CEO options represent increased risk-taking incentives. During normal periods, suppliers reduce RSI to lower their exposure of the increased risk. However, when option grants accompany a shock, suppliers increase RSI because they too have an incentive to increase risk during transition periods. This result shares a similar intuition with Kale et al. (2013) who show that warranty offering firms increase financial risk (through higher debt levels) when distressed but lower debt in non-distress periods.

In the fourth specification, which we use as the baseline model in future tests, we include *Supplier*

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Shock as an additional independent variable. The coefficient estimates and the statistical significance of the three main customer related variables, *Option Grants*, *Cust. Shock*, and *Option Grants* × *Cust. Shock* remain virtually identical to the previous specification. The estimate on *Supplier Shock* is positive (= 0.0025) and statistically significant at the 5% level (t-stat = 2.08). When suppliers experience a shock to their own industry, they also increase RSI. In a sense, this is consistent with risk-taking. Together, the finding that suppliers on average exhibit a negative response to new option grants but a positive response when option grants accompany a shock is consistent with the risk-taking hypotheses.

4.2. Difference-in-Difference Estimation

The results thus far suggest a positive correlation between CEOs' stock and option grants and subsequent suppliers' RSI in periods of shock, but endogeneity presents a challenge, because empirical measures of risk-taking and option pay may be jointly determined. One reason for this is that CEO compensation, supplier selection, and other important corporate decisions are made simultaneously. Consistent with this, Coles et al. (2006) use a simultaneous equation approach to show that the relation between risk-taking and option grants goes both ways, reinforcing the importance of identifying exogenous changes to further establish the link. Another possibility is that firms with different levels of option grants could systematically differ in other aspects that account for the variation in their suppliers' RSI. A third possibility is that a customer firm with risky R&D investments may choose to award more options to its CEO as well as seek high R&D suppliers. This may be the case, especially during transition periods. Finally, there may be time-varying unobservable factors such as CEO risk aversion and changes in investment opportunity or product technology that affect both compensation policy and supplier choice. We use the following specification to estimate a difference-in-difference regression to address the endogeneity issue.

$$\begin{aligned} Supp.\,RSI_{t,t+1} = {} & B_0 + B_1 Treated + B_2 Post + B_3 Cust.\,Shock \\ & + B_4 Cust.\,Shock \times Post + B_5 Post \\ & \times Treated + B_6 Cust.\,Shock \times Treated \\ & + B_7 Cust.\,Shock \times Post \times Treated \\ & + B_8 Supp.\,Shock + B_9 Cust.\,Controls \\ & + B_{10} Supp.\,Controls + C\text{-}S\ FE \\ & + Year\ FE + \mathrm{e}_t \end{aligned} \tag{5}$$

We attempt to overcome these endogeneity issues with two quasi-natural experiments. Specifically, we use the *FAS-123R* mandate as the main experiment and supplement this with the *Sarbanes–Oxley Act* as an additional test. The idea behind these experiments is that an exogenous event such as a policy change or new regulation is not a decision that is under a firms' control. Further, while some firms may be affected (treated group) by this "exogenous" change, others (control group) may not. Once firms are assigned to treated and control groups, we can compare the difference in the outcome variable between the treated and control groups in the post event (policy) to the corresponding difference prior to the event. The difference-in-difference estimator can then be used to interpret the effect of the policy on the treated group relative to the control group. The procedure ensures that time-invariant differences, as well as shared trends between the treated and control groups are differenced away (Roberts and Whited 2012). Recall that we conduct all our analysis using C-S pair fixed effects, which control for time-invariant pair level unobservable factors.

4.2.1. Quasi-Natural Experiment Using FAS-123R.

Our main test employs the 2005 mandate that firms comply with FAS-123R, which calls for increased disclosure on option expensing. Following the effective date for compliance (fiscal years beginning after June 15, 2005 for public firms) resulted in a significant cutback in option grants. Brown and Lee (2011) document a 36% drop in option issuance for the median firm post FAS-123R, while Hayes et al. (2012) show that this decline resulted in a significant reduction in vega (sensitivity of the CEO's firm-specific equity to stock volatility). The authors argue that a lower vega indicates that firms did not maintain the pre-FAS-123R convexity in CEO compensation contracts following the adoption of FAS-123R. As a result, this will cause an (exogenous) decline in the risk-taking incentives for CEOs. This sharp change in the compensation mix, and especially the drop in option grants that arises from FAS-123R, potentially changes the risk-taking incentives for customer CEOs but does not alter supplier RSI. In other words, this event serves as an exogenous shock to firms' willingness to compensate managers with stock options and more importantly, it is arguably exogenous to suppliers' RSI. This makes the implementation of FAS-123R a reasonable choice as a quasi-natural experiment to test the risk-taking hypothesis and allows us to further identify the effect of option grants on suppliers' RSI.

To obtain the control sample, we select customer firms that *did not grant any options* to their CEO in the pre-treatment period (e.g., Bakke et al. 2016, Hayes et al. 2012). We use the two 4-year windows from 2002 to 2005 as the pre- and 2007–2010 as the post-treatment periods, respectively.[3] This provides us with a set of firms whose CEOs are not impacted by

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the large reductions in vega induced by FAS-123R. All the other firms are in the treated group. We then define the binary indicator variables *Treated-FAS* which equals 1 (0) if the customer firm is in the treated (control) group and *Post-FAS* which equals 1 (0) in the post-treatment (pre-treatment) period. Note that a customer firm will be classified as treated or control in *both* the pre- and post-treatment periods. This implies that each C-S pair is also classified as treated or control in both periods. Consequently, any regression that includes C-S pair FE will automatically be estimated without the *Treated* variable, because the main effect of *Treated* is subsumed in the C-S pair FE (see legend in Table 2 of the online appendix to Hayes et al. 2012). Likewise, in the presence of year FE, the variable *Post-FAS* will also be dropped from the estimation because it is subsumed by the year FE (see Table 7 in Bakke et al. 2016). As with the baseline tests, all specifications are estimated with C-S-year as the unit of analysis, include C-S pair FE, and clustering of standard errors at the customer-firm and year levels.

One issue that arises with regard to the difference-in-difference methodology is the power of the test that stems from the underlying assumption that all firms in the treated group are in fact affected by the policy shock (FAS-123R) and have reduced their option grants. Indeed, as Bertrand et al. (2004) contend, the truly affected group is "only a fraction of the treatment group" which can result in higher *t*-statistics. In other words, one would tend to over-reject the null that the FAS-123R policy shock has no impact on option grants. We address this issue in two ways. First, in un-tabulated results, we find that average value of option grants in the post-FAS period is 0.242, which is significantly lower than the corresponding pre-FAS-123R value of 0.417, for all firms in the treated group. Further, of the 196 unique customer firms that appear in both the pre- and post-FAS-123R periods, 150 firms (77%) had a lower average option grants in the post-FAS-123R period *vis-à-vis* the pre-FAS-123R period. For these firms, the average value of *Option Grants* declined significantly from 0.451 in the pre-FAS-123R period to 0.196 in the post-FAS-123R period. For the remaining 46 firms, *Option Grants* increased from 0.294 to 0.428 in the pre- to the post-FAS-123R periods, respectively. Thus, a substantial number of treated firms were in fact truly affected by the policy change. Finally, we follow Bertrand et al. (2004) to mitigate the effect of potentially higher *t*-statistics, by block bootstrapping the standard errors, which they contend is one of the solutions when the number of groups, relative to time periods is sufficiently large—which is the case in our sample. Our results remain robust to this alternate method of computing standard errors.[4]

All specifications are estimated using Equation (5) and include the same set of customer characteristics and supplier characteristics (and supplier shock)-based controls as in column 4 of Table 2 but are suppressed to conserve space. As with the main tests, we include C-S pair fixed effects and cluster all standard errors at the customer-firm and year levels. The results are presented in Table 3. We begin with two-way interactions that do not distinguish between normal and transition periods, and report these in the first two columns in Table 3. In these baseline DID specifications, we exclude the customer shock variable and its interaction with any other variable. The main variable of interest here is the interaction term *Post-FAS* × *Treated-FAS*. The first column shows that the interaction term is positive (=0.0031) statistically significant at the 5% level (*t*-stat = 1.96). The second column replicates the previous specification with the addition of year fixed effects, which subsumes the estimate on *Post*. As before, the interaction is positive (=0.0029) and statistically significant (*t*-stat = 1.82). Together these findings support the risk-taking hypothesis by showing that firms that experience a decline in option grants see a corresponding increase in supplier RSI.

In columns 3 and 4 in Table 3, we present our findings from the FAS-123R natural experiment in the presence of a shock to the customer firm. The third column does not include year FE. We find that there is a reduction in supplier RSI in the post-FAS period—the coefficient on Post-FAS = −0.0060 (*t*-stat = −1.74). More importantly, the coefficient of interest is the triple interaction term *Cust. Shock* × *Post-FAS* × *Treated*, which as expected, is negative (−0.0284) and significant at the 1% level (*t*-stat = −2.60). This shows that an exogenous reduction in option grants (*Post-FAS* = 1) coupled with a shock to the customer (*Cust. Shock* = 1) causes a reduction in suppliers RSI for the treated group (*Treated* = 1) relative to the control group. One could argue that even within the pre- and post-treatment periods, there may be temporal variations in the environment. To address this issue, we add year FE to the specification and report these in column 4. As noted above, the variable *Post-FAS* is dropped (along with *Treated*) from this specification but not any of its interactions with other variables. The results from this test are nearly identical to the earlier findings. In sum, firms with CEOs that receive a larger cut in their stock and option grants during economic shocks see a reduction in their suppliers' RSIs. This finding is consistent with that of suppliers "gambling" their way out of trouble by increasing RSI and further supports our baseline OLS tests.

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Table 3 Natural Experiment using Passage of FAS-123R

Dependent variable	(1) *Supp. RSI*$_{t,t+1}$	(2) *Supp. RSI*$_{t,t+1}$	(3) *Supp. RSI*$_{t,t+1}$	(4) *Supp. RSI*$_{t,t+1}$	(5) *Supp. RSI*$_{t,t+1}$	(6) *Supp. RSI*$_{t,t+1}$	(7) *Supp. RSI*$_{t,t+1}$	(8) *Supp. RSI*$_{t,t+1}$	(9) *Supp. RSI*$_{t,t+1}$	(10) *Supp. RSI*$_{t,t+1}$
	Double difference regressions		Triple difference regressions		Placebo/falsification test 1		Placebo/falsification test 2		Placebo/falsification test 3	
	Using true FAS event				Randomize FAS event ($t-5$)		Randomize FAS event ($t+2$)		Randomize control sample	
Treated-FAS	(*Omitted due to C-S pair fixed effects in all specifications*)									
Post-FAS	−0.0060***		−0.0060*		−0.0013		0.0014		0.0005	
	(−2.83)		(−1.74)		(−0.49)		(0.54)		(0.22)	
Post-FAS × *Treated-FAS*	0.0031**	0.0029*	0.0049	0.0048	0.0015	0.0010	−0.0030	−0.0029	−0.0012	−0.0011
	(1.96)	(1.82)	(1.48)	(1.43)	(0.53)	(0.35)	(−1.21)	(−1.14)	(−0.55)	(−0.47)
Cust. Shock			−0.0027	−0.0025	0.0120	0.0122	−0.0024	−0.0020	−0.0043	−0.0032
			(−0.50)	(−0.46)	(1.25)	(1.26)	(−0.65)	(−0.54)	(−1.10)	(−0.77)
Cust. Shock × *Post-FAS*			0.0069	0.0049	−0.0157	−0.0155	−0.0003	−0.0023	0.0037	0.0030
			(0.85)	(0.60)	(−1.33)	(−1.28)	(−0.05)	(−0.40)	(1.03)	(0.84)
Cust. Shock × *Treated-FAS*			0.0106	0.0101	−0.0116	−0.0126	0.0039	0.0036	0.0029	0.0026
			(1.58)	(1.52)	(−1.19)	(−1.26)	(0.74)	(0.68)	(0.41)	(0.37)
Cust. Shock × *Post-FAS* × *Treated-FAS*			−0.0284***	−0.0272**	0.0199	0.0197	−0.0080	−0.0067	−0.0092	−0.0088
			(−2.60)	(−2.52)	(1.55)	(1.50)	(−0.95)	(−0.82)	(−0.85)	(−0.82)
Customer characteristics			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Supplier characteristics			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Customer–supplier pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.810	0.810	0.807	0.807	0.801	0.801	0.852	0.852	0.877	0.878
Observations	7,604	7,604	7,604	7,604	7,009	7,009	7,172	7,172	3,915	3,915

Notes: The table reports results from difference–in-difference (DID) regressions to investigate the effect of FAS-123R on *Supplier RSI*, using Equation (5). The first four columns report regressions based on the true event date (2006) for the mandate. Columns (1) and (2) establish the baseline case without any interactions with *Cust. Shock*. Columns (3) and (4) are triple difference regressions that replicate the estimations in columns (1) and (2), respectively, but also include interaction terms with *Cust. Shock*. The first placebo experiment reported in columns (5) and (6) replicates the two triple difference regressions by using 2001 ($t-5$) as the false event. The second placebo experiment reported in columns (7) and (8) also replicate the triple difference regressions by using 2008 ($t+2$) as the false event. Both experiments use the same treated and control firms as the true experiment. The third experiment reported in columns (9) and (10) redefines control firms (non-option granting firms in the entire pre-treatment period) with 2006–2007 as the pre-treatment period. All regressions include C-S pair fixed effects. An unreported constant, customer- and supplier-characteristics are included in all specifications. The *t*-statistics are based on standard errors clustered by customer firm and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in the *Data Appendix*.

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4.2.2. Parallel Trends Assumption and Placebo Tests (FAS-123R). A key identifying assumption underlying the DID estimation is that the parallel trend assumption is satisfied, i.e., absent treatment, both the treated and control groups should experience a parallel trend in the outcome variable. In other words, if FAS-123R did not occur the estimated coefficient on the interaction term *Cust. Shock* × *Post-FAS* × *Treated* would not be statistically different from zero. We perform a variety of tests suggested by Roberts and Whited (2012) to check whether the parallel trends assumption is satisfied in our sample. Specifically, we conduct several falsification (placebo) tests to ensure the validity of the above findings and better assess that our results are unique to FAS-123R. These tests are similar to those carried out in Bakke et al. (2016) who study the effect of option pay on risk management using a sample of firms in the oil and natural gas industry.

In the first of three such experiments, reported in columns 5 and 6 of Table 3, we randomize the event. Specifically, we falsify the occurrence of the treatment event to 5 years prior to 2006 and use 1997–2000 and 2002–2005 as the pre- and post-treatment periods. In the second falsification experiment (columns 7 and 8), we move the treatment event 2 years forward to 2008. Here, we use 2004–2007 and 2009–2012 as the pre- and post-treatment periods. In both experiments, the set of treated and control firms are the same as those in the real test. We then re-estimate the two specifications reported in columns 3 and 4. The estimated coefficient on the triple interaction term (treatment effect) is statistically insignificant in all four specifications. The third experiment randomizes the control sample by defining the control group in a placebo period. For this test we use 2006–2007 and 2009–2010 as the pre- and post-treatment periods and replicate the original true test. These results are reported in the last two columns of Table 3. As with the first two placebo tests the coefficient on the triple interaction term is not significant.[5] Overall, these tests support our main findings and further identify the relation between option grants to customer CEOs in periods of shock and suppliers' RSI.

4.2.3. Quasi-natural Experiment Using the Sarbanes–Oxley Act (2002). Our second natural experiment employs the passage of the Sarbanes–Oxley Act (SOX) in 2002 as the natural experiment. One of the main provision of the Act mandates that all Boards have a majority of independent members and fully independent audit, compensation, and nominating committees. While this provision does not directly address a firm's compensation policy, the extant literature provides some evidence that SOX has had an effect on executive compensation. For instance, Cohen et al. (2013) document a significant decline in performance- and incentive-based compensation awarded to CEOs. They argue that this reduction in incentive awards is in part, related to the higher personal costs to executives (from increased criminal and civil penalties and broader financial reporting responsibilities) undertaking risky activities making them more risk averse in the post-SOX period. The authors further contend that rational boards may anticipate the direct effect of SOX on their executives and introduce executive compensation plans that are modified to reflect the reduction in risk-taking propensity. Consistent with their prediction, Cohen et al. (2013) find a significant reduction in incentive-based compensation in the post-SOX period —specifically with respect to options. They report an average annual drop of 6.2% in option compensation (as a proportion of total compensation) following the implementation of SOX. In a related study, Banerjee et al. (2015) find that overconfident CEOs reduce risk following the implementation of SOX and attribute this to improved internal governance and disclosure as a means to restrain overconfident CEOs. While this study does not report any direct effect on compensation they document a nearly 50% decline in the level of idiosyncratic risk following SOX, for firms affected by SOX relative to otherwise comparable firms.

We acknowledge that the provisions in SOX are arguably not a direct "exogenous shock" to compensation policy, as is the case with FAS-123R. Nevertheless, we believe that the above studies make a reasonable case for the use of SOX as a secondary natural experiment in our setting. At the very least, it may help validate our earlier results with FAS-123R as the natural experiment.

We exploit a similar DID methodology to compare the observed effect before and after SOX. First, we categorize firms into treated and control groups as follows. We define the pre- (1996–2001) and post-SOX periods (2003–2006) and restrict our sample to firms that exist for at least 2 years in both periods. We follow Chhaochharia and Grinstein (2009) and define compliance with SOX if firms satisfy the following two conditions *in the pre-SOX period*: (i) have a majority of independent board members, and (ii) have a completely independent audit, compensation, and nominating committees. This is a more stringent criterion for selecting the control group than the one used by Banerjee et al. (2015) who use only the board and audit committee requirements. The median value of firm-years that are compliant in the post-SOX period is 66.66%. We use this as a benchmark and classify a firm as pre-SOX (post-SOX) compliant if it meets the above two conditions at least 66.66% of the time it appears in the pre-SOX (post-SOX) period. The control group consists of firms that are compliant in the

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pre- *and* post-SOX periods. The treated group is the set of firms that are *not* compliant in the pre-SOX period but are compliant in the post-SOX period. There are 655 observations in the control group and 5,750 observations in the treated group. As with FAS-123R, we estimate Equation (5) at the C-S-year level with C-S pair FE in all specifications.

Table 4 reports results from the experiment. We find that the estimated coefficient on the triple interaction term *Cust. Shock* × *Post-SOX* × *Treated-SOX* in the first column is negative (−0.0394) and significant at the 10% level (t-stat = −1.81). In the second specification, which includes year FE, the coefficient estimate continues to remain negative (−0.0428) and statistically significant at the 5% level (t-stat = −1.97). These results reaffirm our earlier findings and show that an exogenous decline in option grants when combined with shock to the customer results in suppliers' reducing their RSI.

4.2.4. Placebo Tests (SOX). The next six columns of Table 4 report results from three falsifications tests to ensure the validity of the above findings and better assess that our results are unique to SOX. These tests are identical to those carried out with FAS-123R as the natural experiment. In the first of three such experiments, reported in columns 3 and 4 of Table 4, we randomize the event. Specifically, we falsify the occurrence of the treatment event to 1999, which is 3 years prior to the actual event and use 1996–1998 and 2000–2002 as the pre- and post-treatment periods. In the second falsification experiment (columns 5 and 6), we move the treatment event 2 years forward to 2004. Here, we use the periods 2001–2003 and 2005–2006 as the pre- and post-treatment periods (we end in 2006 to avoid effects from FAS-123R). In both experiments, the set of treated and control firms are the same as those in the real test. We then re-estimate the two specifications. The estimated coefficient on the triple interaction term (or the treatment effect) is statistically insignificant in all four specifications. The third experiment randomizes the control sample for the second placebo period (second experiment) and replicates the original true test. These results are reported in the last two columns of Table 4. As with the first two placebo tests, the coefficient on the triple interaction term is not significant. Overall, these findings reinforce results from the FAS-123R tests.

5. Additional Analyses and Alternate Explanations

In this section, we conduct several tests to explore alternate hypotheses and reaffirm the validity of our results with cross-sectional and robustness checks.

5.1. Decomposing Shocks

Table 5 presents results from reestimating the baseline specification (column 4 in Table 2). The first three columns are based on productivity shocks followed by demand shocks in the next three. The first column in the table is a subset of firms that experience either a positive productivity shock or no shock. As before, the coefficient on *Option Grants* is negative (−0.0052) and statistically significant (t-stat = −3.56) at the 1% level, consistent with suppliers on average lowering RSI with increasing option grants. In other words, despite a positive shock to the customer, the negative relation between option grants and suppliers' RSI during normal periods continues to hold. The estimate on the interaction term *Option Grants* × *Pos. Shock*, however, is not significant. On the other hand, the interaction term *Option Grants* × *Neg. Shock* in the second specification is positive (0.0071) and statistically significant at the 5% level (t-stat = 2.24). This shows that option grants, when accompanied by a negative productivity shock, lead to higher supplier RSI, consistent with risk-taking. We report sums of the coefficient estimates on *Option Grants*, *Positive-* or *Negative-Shock*, and their corresponding interactions, in the bottom panel of Table 5. None of these sum of coefficients is statistically significant. This implies that the positive effect of the interaction term is offset by the negative effect of the option grants, thereby resulting in no net change in supplier RSI. This finding is again consistent with risk-taking because it suggests that firms issue just enough option grants to offset the potential effect of increased supplier risk-taking from a negative shock. The third specification includes both positive and non-positive productivity shocks. Again the interaction of *Option Grants* with the positive productivity shock is not significant while its interaction with negative shocks remains positive (0.0068) and significant at the 5% level (t-stat = 2.20).

The next three specifications examine positive and negative demand shocks. These results are nearly identical to those with productivity shocks discussed above. The interaction term of *Option Grants* with the *Negative Shock* is positive and significant in both specifications (models 5 and 6). Overall, these results provide further evidence in support of risk-taking. Put simply, when customers experience a negative shock to their industry and increase option grants, potentially amplifying their risk-taking incentives, suppliers follow suit with an increased investment in risky RSI.

5.2. Subsample Analyses, Alternate Explanations, Placebo, and Robustness Tests

Thus far, we have shown that an increase in risk-taking incentive compensation to customer CEOs leads suppliers to increase their relation-specific

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Table 4 Natural Experiment using Passage of Sarbanes–Oxley (SOX)

	Triple difference regressions		Placebo/falsification test 1		Placebo/falsification test 2		Placebo/falsification test 3	
	Using true sox event		Randomize sox event ($t-5$)		Randomize sox event ($t+2$)		Randomize control sample	
	(1) *Supp. RSI*$_{t,t+1}$	(2) *Supp. RSI*$_{t,t+1}$	(3) *Supp. RSI*$_{t,t+1}$	(4) *Supp. RSI*$_{t,t+1}$	(5) *Supp. RSI*$_{t,t+1}$	(6) *Supp. RSI*$_{t,t+1}$	(7) *Supp. RSI*$_{t,t+1}$	(8) *Supp. RSI*$_{t,t+1}$
Treated-SOX	*(Omitted due to C-S pair fixed effects in all specifications)*							
Post-SOX	−0.0012 (−0.22)		0.0028 (1.13)		−0.0007 (−0.09)		0.0016 (0.32)	
Cust. Shock	−0.0241** (−1.99)	−0.0258** (−2.12)	0.0005 (0.08)	0.0001 (0.01)	−0.0147 (−1.02)	−0.0152 (−1.05)	−0.0048 (−0.62)	−0.0062 (−0.82)
Cust. Shock × Post-SOX	0.0319 (1.53)	0.0333 (1.62)	−0.0137*** (−2.58)	−0.0127** (−2.31)	0.0317 (0.64)	0.0336 (0.68)	−0.0006 (−0.05)	0.0021 (0.19)
Post-SOX × Treated-SOX	0.0012 (0.21)	0.0019 (0.34)	−0.0017 (−0.86)	−0.0007 (−0.32)	0.0069 (0.89)	0.0068 (0.89)	0.0003 (0.09)	0.0002 (0.06)
Cust. Shock × Treated-SOX	0.0270** (2.18)	0.0292** (2.35)	0.0048 (0.67)	0.0059 (0.89)	0.0169 (1.16)	0.0168 (1.15)	0.0053 (0.53)	0.0041 (0.40)
Cust. Shock × Post-SOX × Treated-SOX	−0.0394* (−1.81)	−0.0428** (−1.97)	0.0046 (0.68)	0.0041 (0.60)	−0.0365 (−0.73)	−0.0365 (−0.74)	0.0078 (0.54)	0.0086 (0.59)
Customer characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Supplier characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Customer–supplier pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.812	0.813	0.839	0.839	0.826	0.826	0.822	0.823
Observations	6,405	6,405	3,798	3,798	3,446	3,446	2,936	2,936

Notes: The table reports results from difference-in-difference (DID) regressions to investigate the effect of SOX on *Supplier RSI*, using Equation (5). The first two columns report regressions based on the true event date (2002) for the mandate. The first placebo experiment reported in Models 3 and 4 uses 1999 (t–3) as the false event. The second placebo experiment reported in Models 5 and 6 uses 2004 ($t+2$) as the false event. Both experiments use the same treated and control firms as the true experiment. The third experiment reported in Models 7 and 8 redefines control firms (Compliance with SOX in the pre-treatment period) with 2001–2003 as the pre-treatment period ($t+2$ as the false event). All regressions include C-S pair fixed effects. The t-statistics are based on standard errors clustered by customer-firm and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in the *Data Appendix*.

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Table 5 Decomposing Shocks into Productivity and Demand and Positive vs. Negative Shocks

	Productivity shocks			Demand shocks		
	(1) *Supp. RSI*$_{t,t+1}$	(2) *Supp. RSI*$_{t,t+1}$	(3) *Supp. RSI*$_{t,t+1}$	(4) *Supp. RSI*$_{t,t+1}$	(5) *Supp. RSI*$_{t,t+1}$	(6) *Supp. RSI*$_{t,t+1}$
Option grants (β_1)	−0.0052*** (−3.56)	−0.0060*** (−4.13)	−0.0061*** (−4.17)	−0.0051*** (−3.59)	−0.0061*** (−4.06)	−0.0061*** (−4.19)
Positive shock (β_2)	−0.0070 (−1.61)		−0.0099** (−2.30)	−0.0023 (−0.80)		−0.0025 (−1.09)
Opt. grants × Pos. shock (β_3)	0.0076 (0.84)		0.0127 (1.59)	0.0055 (1.02)		0.0080* (1.77)
Negative shock (β_4)		−0.0028* (−1.77)	−0.0029* (−1.85)		−0.0034** (−1.99)	−0.0037** (−2.21)
Opt. grants × Neg. shock (β_5)		0.0071** (2.24)	0.0068** (2.20)		0.0064* (1.89)	0.0066** (2.00)
Supplier shock	0.0028** (2.38)	0.0023* (1.89)	0.0025** (2.15)	0.0025** (2.30)	0.0023* (1.82)	0.0025** (2.10)
New CEO	0.0010 (1.00)	−0.0001 (−0.12)	0.0003 (0.24)	0.0010 (1.34)	0.0003 (0.27)	0.0004 (0.37)
Chairman	0.0000 (0.00)	−0.0017* (−1.70)	−0.0013 (−1.20)	−0.0002 (−0.22)	−0.0011 (−1.06)	−0.0013 (−1.21)
Delta	−0.0000 (−0.15)	−0.0002 (−0.66)	−0.0003 (−1.20)	−0.0001 (−0.24)	−0.0002 (−0.84)	−0.0003 (−1.15)
Old format	0.0017 (0.86)	0.0008 (0.36)	0.0006 (0.27)	0.0017 (0.90)	0.0003 (0.13)	0.0005 (0.22)
Customer size	0.0023 (1.32)	0.0023 (1.27)	0.0018 (0.96)	0.0022 (1.54)	0.0017 (0.88)	0.0016 (0.89)
Customer book leverage	0.0054 (1.08)	0.0098** (1.99)	0.0062 (1.21)	0.0067 (1.52)	0.0068 (1.29)	0.0060 (1.16)
Customer book to market	−0.0005 (−0.20)	0.0017 (0.69)	−0.0004 (−0.15)	0.0011 (0.42)	−0.0007 (−0.25)	0.0001 (0.04)
Customer ROA	0.0139* (1.65)	0.0221** (2.30)	0.0173* (1.73)	0.0146** (1.99)	0.0153 (1.49)	0.0174* (1.75)
Customer SD returns	0.0019 (0.23)	−0.0063 (−0.72)	0.0015 (0.16)	−0.0068 (−0.94)	0.0015 (0.16)	0.0016 (0.18)
Customer R&D	0.1025 (1.57)	0.0750 (1.50)	0.0780 (1.50)	0.1126** (2.08)	0.0570 (0.99)	0.0753 (1.45)
Missing R&D	−0.0035** (−2.52)	−0.0043*** (−3.49)	−0.0045*** (−3.64)	−0.0030** (−2.35)	−0.0044*** (−3.53)	−0.0044*** (−3.55)
Customer ind. HHI	0.0128*** (2.61)	0.0117*** (2.61)	0.0118*** (2.62)	0.0147*** (2.93)	0.0110** (2.40)	0.0113** (2.51)
Supplier size	−0.0052*** (−3.55)	−0.0041*** (−2.98)	−0.0038*** (−2.69)	−0.0040*** (−2.79)	−0.0041*** (−2.75)	−0.0038*** (−2.67)
Supplier book leverage	−0.0021 (−0.74)	−0.0016 (−0.48)	−0.0022 (−0.63)	−0.0021 (−0.78)	−0.0019 (−0.51)	−0.0023 (−0.65)
Supplier ROA	−0.0199*** (−5.34)	−0.0230*** (−6.52)	−0.0234*** (−6.60)	−0.0205*** (−5.59)	−0.0236*** (−6.41)	−0.0234*** (−6.62)
Supplier Ind. HHI	0.0014 (0.57)	−0.0017 (−0.54)	−0.0014 (−0.43)	0.0027 (1.12)	−0.0024 (−0.75)	−0.0014 (−0.43)
Single Cust. Ind. Dummy	0.0017 (1.51)	0.0029** (2.57)	0.0031*** (2.76)	0.0013 (1.08)	0.0035*** (3.13)	0.0031*** (2.80)
$\beta_1 + \beta_2 + \beta_3$	−0.0047 (−0.75)		−0.0033 (−0.66)	−0.0019 (−0.49)		−0.0006 (−0.18)
$\beta_1 + \beta_4 + \beta_5$		−0.0018 (−0.80)	−0.0022 (−1.01)		−0.0030 (−1.26)	−0.0032 (−1.36)
C-S pair FE/year FE	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes	Yes/Yes
Adjusted R^2	0.840	0.817	0.808	0.834	0.807	0.808
Observations	13,876	16,646	17,553	13,492	17,030	17,553

Notes: The table reports results from decomposing the *Customer Shock* variable into *Demand* and *Productivity* Shocks. Within each category, we define a positive (negative) shock if the median industry change for all underlying factors is positive (negative). The first column under each category compares positive shocks with no shock. The second compares negative shocks with no shock. The third column includes both positive and negative shocks. The sample consists of 17,553 observations, spanning the 1993–2013 period. The dependent variable is *Supplier RSI*. The bottom panel presents tests for the sum and significance of relevant coefficients. All regressions include C-S pair FE, year FE, and a constant. The *t*-statistics are based on standard errors clustered by customer-firm and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in the *Data Appendix*.

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investments, consistent with the argument that the entire supply chain has an incentive to increase risk when the customer firm increases risk-taking during a shock. In this section, we discuss scenarios that are likely to amplify or attenuate the relation between customer risk-taking and supplier RSI. We also consider several alternate explanations that may potentially explain our findings. Lastly, we conduct several placebo and robustness checks to further strengthen our inferences. To this end, we re-estimate the main baseline specification (Equation (4); column 4 in Table 2) on specific subsamples of the data. These results are reported in Table 6 where each row corresponds to a different subsample. For exposition and to conserve space we only tabulate estimates for the three main variables of interest: *Option Grants*, *Cust. Shock*, and *Option Grants* × *Cust. Shock*. All specifications are estimated exactly as in the baseline case (with *Supplier Shock*, all *Customer-* and *Supplier-characteristics*, C-S pair FE, year FE, and clustering of standard errors by customer-firm and year).

5.2.1. Cross-sectional Analyses.

Financial Constraintsand Distress: Arguably, a customer firm that is financially constrained may seek to gamble its way out of the situation and thus incentivize its managers with a greater proportion of option pay. Suppliers can choose to either go along with this strategy to potentially share the increase in *expected payoff* or, reduce RSI to avoid increased exposure to a financially constrained customer. We separate the sample into financially constrained and unconstrained firms based on the Hadlock and Pierce (2010) measure (see *Data Appendix*). Panel A1 in Table 6 shows results from several slices of the full sample along this dimension. From the first two rows, we find that regardless of whether customers are financially constrained, suppliers on average reduce their RSI when customers' CEOs receive option grants. However, they increase RSI (as in the main baseline result) when such grants are accompanied by a shock, but only when customers are financially constrained (estimate = 0.0104; *t*-stat = 2.26). We find a similar pattern (reduce RSI on average but increase RSI when option grants accompany a shock) when the (i) supplier firm is itself financially constrained (row 3) and (ii) when both firms are financially constrained (row 5).

In a similar vein, we partition the sample based on the Altman (1968) *Z*-score (see *Data Appendix*), a measure indicating economic distress and likelihood of liquidation. These results are reported in Panel A2 of Table 6. As with financial constraints, there is an average negative relation between customers' option grants and suppliers' RSI regardless of whether customers are in distress, when the supplier firm itself is in distress, or both are in distress, as seen from the negative and significant estimates on *Option Grants* in the corresponding rows. Unlike the scenario with financial constraints, however, suppliers increase RSI when options accompany a shock, and the customer is *not* in distress. Analogous to the financial constraints scenario, when the supplier firm is itself distressed, the interaction term is positive (=0.0112) and statistically significant at the 1% level (*t*-stat = 2.99). Finally, suppliers increase RSI either when both suppliers and customers risk liquidation (stronger result) or when neither risks liquidation (weaker result) (rows 5 and 6). In summary, these results are consistent with the argument that suppliers are willing to bear increased risk (through higher RSI) when customers increase risk and are financially constrained, but not when their customers are in economic distress and risk liquidation.

Ex ante Characteristics of Customer- and Supplier-firms: Next, we explore scenarios based on *ex ante* characteristics of customer and supplier firms that likely amplify or attenuate the observed effect. First, firms with higher growth options (proxied by Tobin's *Q*) are more likely to use option grants (equity grants in general) because it is costlier to monitor managers in riskier environments than those with poor growth opportunities (e.g., Gaver and Gaver 1993, Smith and Watts 1992). Similarly, customers operating in a less competitive environment (proxied by industry HHI index) are more likely to view options as increasing shareholder alignment, than those in competitive industries where rents are lower and option grants may be used to incentivize risk-taking behavior to remain competitive. Panels A3 and A4 of Table 6 presents these findings. First, as with the baseline tests, suppliers on average lower RSI with increasing customer option grants. When combined with an uncertainty shock, however, suppliers increase RSI when customers face poor growth options or are in highly competitive environments. We interpret the increase in RSI during shocks as further evidence in support of risk-taking along the supply chain.

5.2.2. Alternate Explanations.

In the following sub-sections, we explore other arguments besides risk-taking that potentially explain our findings.

Customer R&D Intensity: It could be the case that high R&D supply chains typically use more options in their compensation mix because R&D projects are risky and accordingly require appropriate incentives to preclude firms from sub-optimally investing in R&D (e.g., Cheng 2004). This may be especially important during transition periods when customer firms award higher risk-taking incentives to encourage R&D investments which in turn leads the entire supply chain to increase R&D spending. To test this argument, we partition the sample

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Table 6 Cross-Sectional Analyses, Alternate Explanations, Placebo Tests, and Robustness Checks

Description	*Option Grants*	*Cust. Shock*	*Option Grants × Cust. Shock*	Adj. R^2	Observations
Panel A: Cross-sectional tests					
Panel A1: Financial constraints					
Cust. financially constrained	−0.0051* (−1.95)	−0.0033 (−1.30)	0.0104** (2.26)	0.79	7,031
Cust. financially unconstrained	−0.0065*** (−3.82)	−0.0035* (−1.70)	0.0037 (1.05)	0.83	10,522
Supp. financially constrained	−0.0135*** (−3.88)	−0.0068** (−2.32)	0.0154*** (2.84)	0.78	8,276
Supp. financially unconstrained	−0.0008 (−0.95)	−0.0022* (−1.96)	0.0025 (1.21)	0.84	9,277
Cust. and supp. financially constrained	−0.0102* (−1.87)	−0.0063 (−1.50)	0.0163** (2.11)	0.76	3,680
Cust. and supp. financially unconstrained	−0.0007 (−0.73)	−0.0013 (−0.92)	0.0003 (0.11)	0.83	5,926
Panel A2: Financial distress					
Cust. financially distressed	−0.0070*** (−3.03)	−0.0026 (−1.17)	0.0058 (1.24)	0.79	11,236
Cust. financially not distressed	−0.0036** (−1.97)	−0.0045** (−2.02)	0.0077** (2.49)	0.87	6,317
Supp. financially distressed	−0.0078*** (−3.65)	−0.0054*** (−2.62)	0.0112*** (2.99)	0.81	11,794
Supp. financially not distressed	−0.0014 (−0.90)	−0.0007 (−0.34)	0.0061 (1.41)	0.87	5,759
Cust. and supp. financially distressed	−0.0091*** (−2.72)	−0.0060* (−1.94)	0.0132** (2.19)	0.79	7,514
Cust. and supp. financially not distressed	−0.0012 (−0.74)	−0.0028 (−1.02)	0.0096* (1.84)	0.92	2,037
Panel A3: Ex ante characteristics of customer firms					
Cust. high Tobin's *Q*	−0.0069*** (−2.78)	−0.0012 (−0.44)	0.0039 (0.92)	0.80	9,464
Cust. low Tobin's *Q*	−0.0035** (−2.10)	−0.0058*** (−3.26)	0.0092*** (2.65)	0.84	8,089
Cust. high HHI (less competitive)	−0.0036*** (−3.04)	−0.0025* (−1.87)	0.0042 (1.21)	0.83	10,655
Cust. low HHI (more competitive)	−0.0094** (−2.20)	−0.0065* (−1.86)	0.0127** (2.23)	0.78	6,898
Panel A4: Ex ante characteristics of supplier firms					
Supp. high Tobin's *Q*	−0.0111*** (−3.22)	−0.0085** (−2.57)	0.0137** (2.46)	0.80	7,562
Supp. low Tobin's *Q*	−0.0035** (−2.32)	−0.0003 (−0.16)	0.0035 (1.07)	0.85	9,991
Supp. high HHI (Less Competitive)	−0.0030* (−1.87)	−0.0009 (−0.56)	0.0035 (1.16)	0.82	8,288
Supp. low HHI (More Competitive)	−0.0080*** (−3.12)	−0.0060** (−2.16)	0.0105** (2.07)	0.81	9,265
Panel B: Alternate explanations					
Panel B1: Customer R&D intensity					
Cust. positive R&D	−0.0115*** (−3.89)	−0.0059** (−2.32)	0.0118*** (2.71)	0.79	9,045
Cust. zero R&D	−0.0001 (−0.11)	−0.0016 (−1.40)	0.0018 (0.70)	0.84	8,508
Panel B2: M&A activity					
Exclude cust. involved in M&A	−0.0052*** (−3.75)	−0.0054*** (−2.77)	0.0112*** (2.62)	0.84	12,600
Exclude supp. involved in M&A	−0.0072*** (−4.24)	−0.0035** (−1.99)	0.0067** (2.06)	0.82	14,900
Exclude cust. & supp. involved in M&A	−0.0058*** (−3.57)	−0.0059*** (−2.66)	0.0124** (2.51)	0.85	10,767
Panel B3: Earnings management	*Pred. Earnings Mgmt.*	*Cust. Shock*	*Pred. Earnings Mgmt. × Cust. Shock*	Adj. R^2	Observations
Second stage: effect on RSI	−0.2964** (−2.13)	−0.0018 (−0.59)	0.0094 (0.20)	0.81	16,075

(continued)

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Table 6 ***(Continued)***

	Option Grants	*Cust. Shock*	*Option Grants × Cust. Shock*	Adj. R^2	Observations
Panel C: Placebo tests					
Panel C1: Supp. single vs. multiple industries					
Supp. serves single industry	−0.0063*** (−3.41)	−0.0032* (−1.68)	0.0078** (2.21)	0.81	13,192
Supp. serves multiple industries	−0.0021 (−0.88)	−0.0060* (−1.96)	0.0067 (1.29)	0.77	4,361
Supp. has single cust. per industry	−0.0062*** (−3.33)	−0.0044** (−2.43)	0.0090*** (2.68)	0.83	12,574
Supp. has multiple cust. per industry	−0.0033* (−1.76)	−0.0036 (−1.09)	0.0050 (0.75)	0.80	4,979
Some cust. grant options and some don't	−0.0007 (−0.42)	−0.0033 (−0.90)	0.0018 (0.35)	0.92	1,745
Panel C2: Other placebo tests					
Government as customer	0.0001 (0.05)	0.0073 (1.32)	−0.0005 (−0.07)	0.85	1,432
Family-firm as customer	−0.0023 (−0.59)	0.0009 (0.25)	−0.0188 (−1.28)	0.85	2,775
Family-firm as supplier	0.0015 (0.57)	0.0092 (1.47)	−0.0166 (−1.14)	0.79	1,111
Family-firm as either customer or supplier	−0.0014 (−0.52)	0.0035 (1.03)	−0.0159* (−1.66)	0.83	3,541
Cust. or supp. is utility company	0.0003 (0.36)	0.0001 (0.14)	−0.0007 (−0.57)	0.98	443
Panel D: Robustness checks					
Excluding Delta	−0.0061*** (−4.20)	−0.0036** (−2.37)	0.0072** (2.48)	0.82	17,553
Excluding newly promoted CEOs	−0.0060*** (−3.76)	−0.0047*** (−2.96)	0.0095*** (3.13)	0.81	16,075
Excluding retiring CEOs	−0.0068*** (−4.14)	−0.0036** (−2.25)	0.0058** (1.97)	0.81	16,289
Excluding years with reporting format changes	−0.0063*** (−4.01)	−0.0035** (−2.23)	0.0079*** (2.69)	0.82	16,379
Dist. to default as firm-level shock measure (end of FY)	−0.0045** (−2.40)	−0.0048*** (−2.72)	0.0119** (2.34)	0.84	8,352
Dist. to default as firm-level shock measure (Ann. Ave.)	−0.0048** (−2.53)	−0.0023* (−1.75)	0.0077** (2.18)	0.84	8,352

Notes: The table reports 4 panels where each row of the table estimates the baseline specification in Model 4 of Table 2. Only the three main coefficients of interest are reported. Panel A reports cross-sectional tests based on subsamples of financial constraints, financial distress, and *ex ante* option granting characteristics of customer and supplier firms. Panel B reports tests of alternate explanations that include R&D intensity, merger activity, and earnings management. Panel C reports placebo tests, number of customer industries, government, and family firms as customers. Panel D reports robustness tests that (i) exclude observations potentially affecting option grants, and (ii) use the *Distance to Default* as an alternate shock measure. The sample consists of 17,553 observations, spanning the 1993–2013 period. The dependent variable is *Supplier RSI*. All regressions include supplier shock, customer and supplier controls (exactly as in Model 4 of Table 2), C-S pair FE, year FE, and a constant. The *t*-statistics are based on standard errors clustered by customer-firm and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in the *Data Appendix*.

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based on customers with positive and zero R&D expenditure and report our findings in Panel B1 of Table 6. We find that our baseline results are reflected in the positive R&D subsample which is consistent with the R&D intensive supply chains requiring higher levels of option compensation, and this is even more important when there is an economic shock. We believe that this explanation does not contradict the risk-taking argument and rather suggests that risk-taking is more prevalent in high R&D supply chains.

M&A Activity along the Supply Chain: Harford (2005) shows that industry shocks are correlated with merger waves. It is, therefore, possible that some of the observed effects are due to customer firms engaging in a merger and subsequently adjusting their CEO's compensation. Thus, a merger could affect both CEO grants and suppliers' R&D. To rule out this alternate explanation, we drop firms that are involved in a merger during the sample year. The first row in Panel B2 of Table 6 reports baseline results by excluding all, merger involved customers, followed by suppliers, and finally both. In all three specifications, the sign and statistical significance of the point estimates for all variables of interest are consistent with the main findings. We can, therefore, rule out mergers as a confounding event that drives our results.

Earnings Management: Bergstresser and Philippon (2006) show that firms with higher incentive pay also have higher levels of earnings management. Further, Raman and Shahrur (2008) document that customers' earnings management affects supplier RSI. Taken together, these findings could imply that customer equity grants affect earnings management, which subsequently enhances supplier RSI. To test this argument we use a 2SLS methodology (described in the *Data Appendix*). In the first stage we predict the level of earnings management (measured based on discretionary accruals used by Raman and Shahrur 2008) with option grants and a set of controls. We then compute the predicted value of earnings management (*Pred. Earnings Mgmt.*) and replace *Option Grants* with this predicted value in the second stage. Barring this replacement, the second stage is exactly the same as in Equation (4). The results are tabulated in Panel B3 of Table 6. While there is a negative association between (predicted) earnings management and RSI on average, the interaction term is not statistically significant—ruling this out as an alternate explanation.

Signaling Better Prospects: Consider the premise that customer CEOs could influence their compensation and hold private information (Gao 2015, Inderst and Mueller 2010). Because higher option grants tie the CEO's compensation to the firm's future payoffs, it may convey positive information about the firm's future prospects. Suppliers' who look to the firm for information, especially during transition periods, may read into higher incremental option grants as a positive signal and invest more in RSI to share the higher expected payoffs. The preceding argument however, should also hold, perhaps more so, for higher incremental stock awards made to customer CEOs. In un-tabulated results, we find that this is not the case—incremental stock awards have no effect on suppliers' RSI either during normal- or transition-periods.

5.2.3. Placebo Tests.

Supplying to Single Vs. Multiple Industries: The next test explores the customers' industry distribution from the suppliers' perspective. Specifically, we examine subsets where the reported customers for a supplier are members of the same or multiple industries and tabulate findings in panel C1 of Table 6. The idea behind this classification is that single-industry suppliers are associated with a "clean event," i.e., their customers either experience a shock or they don't. The first row of panel C1 shows that the main results stem from suppliers who supply to a single industry. In contrast, when there is industry-level diversity in their customer portfolio, suppliers do nothing—the estimate on *Option Grants* and its interaction with shock is not statistically significant (row 2). Next, to examine heterogeneity in option grants related to a supplier's customer portfolio, we compare suppliers serving single *vis-à-vis* multiple customers per industry. All three coefficients are statistically significant only for the single customer case (row 3). Suppliers with multiple customers in the same industry decrease RSI with option grants when there is no shock (row 4). However, suppliers do nothing when faced with multiple customers and some issue options while others do not. This scenario behaves as a placebo test and reaffirms our findings.

Government, Family owned, and Utility Firms: We examine other scenarios that are unlikely to have any effect on suppliers RSI. The first involves supplying to the government (local, state, or federal). The dataset identifies the customer only as a government entity and provides the transaction amount. We compute the importance of the government as a customer to the supplier (*Cust. Imp*) as the ratio of the transaction to the supplier's total sales and create a subset where the supplier has a positive sale amount to the government and re-estimate the baseline specification for this subset. Government customers are unlikely to receive performance pay or risk-taking incentives even when they are in distress. Next, we obtain the identities of family-owned customer firms from data made available by Anderson et al. (2012). We then create a subset where the customer firm is family

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owned. The incentives of CEOs in family firms are also likely to be different from those of a traditional public company. Lastly, we create a subset if either the customer or supplier is a utility company (SIC 49). Since these firms are subject to greater regulation and because their customer/supplier base is relatively stable they are unlikely to engage in excessive risk-taking. In all scenarios, we do not expect to find any evidence of risk-taking. Panel C2 of Table 6 shows that this is indeed the case.

5.2.4. Robustness Checks. Finally, we conduct several robustness checks to (i) ensure that our results are not driven by certain specific CEO compensation related statistics, and (ii) use an alternate firm-level measure of shock. We report the three main coefficients of interest for each of these checks in panel D of Table 6. First, we exclude the variable *Delta* as a control from the baseline specification because *Delta* may be correlated with *Option Grants*. Next, we exclude all instances when the CEO is in her first year at the helm—new CEOs often receive a compensation package that may include substantially high bonuses or equity awards. We also drop CEOs above the age of 62 because they are close to retirement and their risk-taking incentives may not depend on their annual compensation. Subsequent to the FAS-123R mandate, there is a change in the reporting format of compensation related variables. To address this, we include the binary indicator *Old Format* in all our specifications. Nevertheless, following the recommendation of Coles et al. (2014) (Appendix A.3.3), we drop the fiscal year 2006 for all firms and 2007 for those firms that continued to report based on the old format in 2006. These results are presented in rows 1–4 of panel D, respectively. Our main results remain robust to all the four exclusions discussed above—the sign and significance of all three coefficients of interest are similar to the baseline case.

The last two rows of panel D present results from an alternate shock measure. We construct the variable *Expected Default Frequency* (or *Distance to Default: EDF*) based on Bharat and Shumway (2008) and Phillips and Sertsios (2013), which represents a firm's default probability. This variable is calculated for each year and month for customer and supplier firms in our sample (detailed in *Data Appendix*). Due to data restrictions, we lose about 50% of the sample. We then replace the variable *Cust. Shock* and *Supp. Shock* with the corresponding *EDF* variables computed in two ways; the first is the value of *EDF* in the last month of the fiscal year and the second is the monthly average over the entire year. These results are shown in the last two rows of panel D. Again, our main conclusions remain robust to this alternate version of shock.

6. Conclusion

We use a comprehensive dataset comprised of 17,553 customer–supplier transactions over the 1993–2013 period to investigate the role of customer firm's CEO incentives in shaping suppliers' RSI, especially when customers experience a shock to their industry. The risk-taking hypothesis suggests that customer firms offer risk-taking incentives to their CEOs during transitional periods, which also induces risky investments from their suppliers—leading to risk-taking along the entire supply chain.

In our baseline analysis, we find that suppliers decrease RSI when customer CEOs receive a new option grant, but increase RSI when such grants are accompanied by a shock to the customer. These results support risk-taking. We use the FAS-123R mandate to further establish the link between customer CEO's risk-taking incentives and suppliers' RSI. We find that the sharp decline in option grants induced by FAS-123R causes a reduction in supplier RSI for firms that are affected by the regulation relative to those that are not. We confirm these findings using SOX as an additional natural experiment. Next, we segregate customer shocks into positive and negative components. We find that suppliers increase RSI when option grants accompany a negative shock, but not when the shock is positive. Together, these results lend additional support for the risk-taking argument. Finally, we conduct several cross-sectional tests to identify scenarios that amplify or attenuate the role of customer option grants and supplier RSI.

One limitation of our study is lack of data on explicit contracts between suppliers and customers. Such explicit data would be valuable in coming up with a more accurate measure of RSI. Another data limitation is the exclusion of suppliers whose sale does not currently meet the 10% threshold required for disclosure, but who make RSIs in anticipation of future business. Nevertheless, as Cen et al. (2016) and Shenoy and Williams (2017) document, this reporting threshold does not appear to be binding. Rather, suppliers appear to report all customers they deem important to their line of business, regardless of whether they are below the 10% threshold. Indeed, approximately 25% of the C–S transactions in our sample fall in this category.

We contribute to the literature on the interplay between corporate policy and product markets as well as to the supply chain literature. In a comprehensive survey of the SCM literature, Kouvelis et al. (2006) recognize the paucity of research that uncovers the hidden information in contracts along the supply chain. Similarly, Radhakrishnan et al. (2014) note that the supply chain management literature has not considered the spillover effects of customers' information quality on supplier behavior. Our study is an attempt

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to fill this gap. More importantly, we identify the effect of an important corporate policy, namely customers' CEO incentives on suppliers' RSI.

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Data Appendix
Variable Definitions and Data Sources

This appendix defines all the variables used in the study. The data items taken from *Compustat* are denoted by their variable names. All returns data are from the *Center for Research in Security Prices* (*CRSP*). Compensation related variables are from *Compustat*'s *ExecuComp*. All variables are defined in the same way for the customer firm and supplier firm (where applicable) and the variable definitions use *Compustat* variable names.

Variable	Definition
Compensation and incentives	
Short-term compensation (ST Comp)	Annual sum of (Salary + Bonus + Other annual payments)
Long-term compensation (LT Comp)	Annual sum of (Restricted stock grants + Black-Scholes value of options granted + Long-term incentive payouts + Total other annual payments)
Total compensation (Tot Comp)	ST Comp + LT Comp
Option grants	Black-Scholes value of all options granted during the year/Tot Comp
Stock awards	Value of stocks awarded during the year/Tot Comp (Annual)
Bonus	Bonus/Tot Comp
Delta	$ Change in the CEO's portfolio of stock and options at the beginning of the year to a 1% change in stock price: (Coles et al. 2006, Core and Guay 2002)
Relation-specific investment (RSI)	
Supp. R&D	Max (XRD,0)/(AT)
Transaction ($ million)	Sale amount in millions by a supplier to firm
Cust. Importance (Cimp)	Transaction/Supplier's Annual Sale (SALE)
Number of suppliers (s)	Number of reported suppliers associated with each customer firm
Supp. RSI$_t$ (C-S pair year)	Cimp × Supp. R&D
Supp. RSI$_{t,t+1}$ (C-S pair year)	*Supp. RSI*$_t$ + *Supp. RSI*$_{t+1}$
Shock related variables	
Principal component	The first principal component from a Principal Component Analysis (PCA) median absolute change in each of the following seven variables for each 2-digit SIC industry-year; Net income to sales (NI/Sale), Asset turnover (Sale/lag(AT)), R&D (XRD/lag(AT)), Capital expenditures (CAPX/lag(AT)), Employee growth (1-year change in EMP), ROA (NI/lag(AT)), and Sales growth in previous year; (Harford 2005)
Customer/supplier shock	Dummy = 1 if PC is in the top quartile and 0 otherwise; (Ahern and Harford 2014)
Demand shock	Follow the same procedure as above, using the following four of the seven variables; R&D, Capital Expenditure, Employee growth, and Sales growth
Productivity shock	Follow the same procedure as above, using the following three of the seven variables; Net income, ROA, and Asset turnover
Positive demand (productivity) shock	If the *raw median changes in* <u>*all*</u> 4 (3) underlying variables used to compute demand (productivity) shocks are <u>*positive*</u>, for each 2-digit SIC industry-year
Negative demand (productivity) shock	All non-positive shocks
Expected default frequency (dist. to default)	Computed at the firm level. Follow the methodology in Bharat and Shumway (2008) on pages 1347–1348. Also used by Phillips and Sertsios (2013). $$DD = \{\ln[(E+F)/F] + (r_{i\ t-1} - 0.5\sigma_V^2)T\}/(\sigma_V\sqrt{T}); \quad EDF = N(-DD),$$ where D = total debt; E = MV of Equity; $F = D + E$; $r_{i\ t-1}$ is the firm's stock returns in the previous year; σ_V is the weighted average volatility of firm's debt and equity. We compute the values of monthly EDFs for firms with available data and compute two measures; (i) fiscal year end EDF and (ii) mean EDF over the year
Natural experiments (FAS-123R and SOX)	
Post-FAS	Dummy = 1 for the Post-FAS period (2007–2010) and 0 for Pre-FAS (2002–2005)
Control-FAS	All firms which <u>*didnot*</u> grant any <u>*options*</u> to its CEO in the entire pre-FAS period
Treated-FAS	All firms not in the Control sample
Post-SOX	Dummy = 1 for the Post-SOX period (2003–2006) and 0 for Pre-SOX (1996–2001)

(continued)

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Appendix ***(Continued)***

Post-SOX	Dummy = 1 for the Post-SOX period (2003–2006) and 0 for Pre-SOX (1996–2001)
Compliant	A firm is SOX compliant in any year if it has a majority of independent members on the Board and independent audit, nominating, and compensation committees
Control-SOX	Firm is compliant in the pre- and post-SOX periods and appears at least twice in both
Treated-SOX	Dummy = 1 for all firms that are not SOX compliant in the pre-SOX but compliant in the post-SOX period and appear at least twice in both periods, 0 for control group
Control variables	
Old format	Dummy = 1 if compensation is reported as per old format, 0 otherwise
Chairman	Dummy = 1 if CEO is also the Chairman of the Board and 0 otherwise
New CEO	Dummy = 1 if CEO is in the first year as CEO and 0 otherwise
Size	Log (Sale)
Leverage	Ratio of long-term debt to Total assets ((DLC + DLTT)/AT)
Book to market	Ratio of total assets to market value of equity and book value of debt (AT/(AT − CEQ + PRCC_F × CSHO)
ROA	Return on Assets (IB/AT)
SD returns	Std. dev. of 12 monthly returns preceding sample year
R&D	Max(XRD,0)/Lag (AT)
Missing R&D	Dummy = 1 for observation where missing value for XRD is replaced with 0 and 0 otherwise
Industry HHI	Annual sales-based Herfindahl index, using all firms in the firm's 3-digit SIC industry
Single industry dummy	Dummy = 1 if all customers in a year are from the same 2-digit SIC code, 0 otherwise
Cross-sectional analyses and others	
Variables used in cross-sectional analyses	
Financial constraint	A firm is *Financially Constrained (Unconstrained)* if its HP index of financial constraints is above (below) the sample median where; $$HP\ Index = -0.737 \times Ln(HPAT) + 0.043 \times \{Ln(HPAT)\}2 - 0.040 \times Firm\ Age$$ *HPAT* = Max(Total Assets, 4500) in 2004 real dollars; *Firm Age* = # of years since first appearance on *Compustat* (Capped at 37 years); (Hadlock and Pierce 2010)
Financial distress	A firm is *Financially Distressed* (*Not Distressed*) if its Altman's *Z*-score is above (below) the sample median, where $$Z\text{-score} = 3.3 \times OIADP/AT + 1.2 \times (ACT - LCT)/AT + Sale/AT + 0.6 \times PRCCF \times CSHO/(DLTT + DLC) + 1.4 \times RE/AT$$
M&A activity	A firm is involved in a merger during the year if it appears either as a target or bidder in *SDC Platinum's M&A Database* with an announcement date in the firm's fiscal year
Earnings management	Following Raman and Shahrur (2008) we compute *Predicted Earnings Management* as the predicted value of discretionary earnings (*DA Hat*), where $$DA\ Hat = TA - NDA\ Hat;$$ *TA* = (Del ACT − Del LCT − Del CHE + Del DLC − Depn.)/Lag(AT); *NDA Hat* = Predicted value of TA (*TA Hat*) from the following regression; $$\text{TA} = \text{B}_1(1/\text{Lag(AT)}) + \text{B}_2(\text{Del Sale} - \text{Del(RECT} - \text{RECD})) + \text{B}_3(\text{PPEGT/Lag(AT)}) + \text{B}_4(\text{ROA}) + \text{B}_5(\text{Book to Mkt.}) + \epsilon;$$ The above specification is estimated without a constant, by 2-digit SIC industry-year.
Family-firm	We identify family firms based on the data provided on Ron Anderson's website based on Anderson et al. (2012, 2009). Source: http://www.ronandersonprofessionalpage.net/data-sets.html
Government as customer	We sum the sales to all customers identified by the *Compustat Segment* files as either a Local, State, or Federal Government entity. We classify all observations with a positive sale to any of these entities as a *Government Customer*
Utility company	We classify a firm as a *Utility* company if its 2-digit SIC = 49

Notes

[1]There are 57 observations where firms name each other as a master customer, such as Pfizer and Cardinal Health. We retain these because they represent actual transactions. Dropping these pairs has virtually no effect on our results.

[2]Specifically, we use the *ExecuComp* variables *option_awards_blk_value* for the old reporting format (pre-2006 fiscal years for most firms) and *option_awards_fv* for the new reporting format. A CEO's total annual compensation is the sum of the following *ExecuComp* variables under the old reporting format (pre-2006): *salary*, *bonus*,

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othann, *rstkgrnt*, *option_awards_blk_value*, *ltip*, *allothtot*. Under the new reporting format, we sum the following variables: *salary*, *bonus*, *noneq_incent*, *option_awards_fv*, *stock_awards_fv*, *defer_rpt_as_comp_tot*, *othcomp*.

[3]Coles et al. (2014) suggest dropping the year 2006 due to reporting format changes (p. 43, Appendix A.3.3).

[4]Specifically, we use C-S pairs as blocks and estimate the standard errors based on 500 replications of the model.

[5]We use a shorter window here to have a reasonable number of firms in the control sample in the pre- and post-treatment periods. There are not enough firms that qualify as a control (never granted any options in the pre-treatment period) and which appear in the post-treatment period if we use a 3-year (or more) period on either side of the placebo event (2008).

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